33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10002

F16-SR-002
415V FLOTATION SWITCHROOM

F16-TR-005
TRANSFORMER ROOM

F16-TR-005
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF

SWR EARTH BAR
Tx EARTH BAR
NOTE 3
4000/ 1A
2.5VA
5P20

F16-SW-005
FLOTATION BLOWERS & COMPRESSORS 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
2.5VA
5P20
4000/ 5A
2.5VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
ACB
AFD
4000A
24VDC CONTROL POWER SUPPLY
NOTE 7
00F-08
125A
160A
SD
NOTE 6

415V 3NE~ 50Hz 4000A 65kA for 1sec
INC

01F-09	03F-09	4F-09	1R-09	3R-09	1F-01	3F-01	4F-01	1R-01	3R-01
800A	800A	800A	800A	xxxA	20A	20A	20A	20A	xxxA
100/1A	100/1A	100/1A	100/1A	xxVA	70A	70A	70A	70A	xxVA
					EOL	EOL	EOL	EOL	EOL

CONTINUED ON SHT 2
2000-F16-SLD-EL-10212

E-SHUT
F16SR002-LCS
NOTE 12

F16-BW-401	F16-BW-402	F16-BW-403	F16-BW-404	F16-BW-405	F16-FN-487	F16-FN-488	F16-FN-489	F16-FN-490	F16-FN-505
315kW	315kW	315kW	315kW	315kW	0.55kW	0.55kW	0.55kW	0.55kW	0.55kW
FLOTATION AIR BLOWER 1	FLOTATION AIR BLOWER 2	FLOTATION AIR BLOWER 3	FLOTATION AIR BLOWER 4	FLOTATION AIR BLOWER 5	FLOTATION AIR BLOWER 1 MOTOR COOLING FAN	FLOTATION AIR BLOWER 2 MOTOR COOLING FAN	FLOTATION AIR BLOWER 3 MOTOR COOLING FAN	FLOTATION AIR BLOWER 4 MOTOR COOLING FAN	FLOTATION AIR BLOWER 5 MOTOR COOLING FAN
2000-F16-SCM-EL-10255	2000-F16-SCM-EL-10256	2000-F16-SCM-EL-10257	2000-F16-SCM-EL-10258	2000-F21-SCM-EL-xxxxx (FUTURE)	2000-F16-SCM-EL-10247	2000-F16-SCM-EL-10248	2000-F16-SCM-EL-10249	2000-F16-SCM-EL-10250	2000-F21-SCM-EL-xxxxx (FUTURE)

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F16-SW-005 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F16-BLK-IC-10001.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

2000-F16-SLD-EL-10211_Rev2

Rougher & scavenger flotation F16-SW-005 415V
MCC single line diagram sheet 1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235I7 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F16 - ROUGHER & SCAVENGER FLOTATION\2000-F16-SLD-EL-10211.DWG 03SEP26 2:29PM ROBIN.LAW